Comparative evaluation of AI-generated image detectors: DDA, GAPL, SPAI, PRL, Community Forensics 384, B-Free

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Abstract

The six detectors are compared on 87 human photographs, their reduced-resolution versions, 87 AI reconstructions and 72 text-generated AI images, with each set evaluated without noise and then with Gaussian noise. The 666 files yield 3,996 results. Across the 333 files in each condition, the overall proportion of correct decisions ranges from 53.8 to 74.2% without noise and from 49.2 to 57.1% with noise. Adding noise increases false alerts on human photographs for five of the six detectors, while detection of AI images decreases for five of them. Preparing the reduced-resolution versions also changes scores and decisions. Performance therefore depends strongly on the sets and transformations studied; its generalisation remains limited by the homogeneity of the photographic corpus and the use of a single generation source.

Conclusion

The observed performance does not identify a detector that is reliable across all the conditions studied. Noise reduces the overall proportion of correct decisions for all six models, and the preparation of human photographs substantially changes the results of some of them.

Model reliability and comparison with chance

The comparison with chance uses balanced accuracy and AUC, whose no-discrimination reference is 50. For each condition, the overall results combine 333 files: 174 human and 159 AI. Always deciding ‘human’ would already yield 52.3% correct decisions; exceeding 50% on this unadjusted measure is therefore insufficient to establish a detector's usefulness.

Without noise, GAPL and DDA achieve balanced accuracies of 73.8 and 73.4, with AUCs of 84.3 and 81.6. SPAI also exceeds the reference of 50 in the observed measurements, with a balanced accuracy of 66.2 and an AUC of 72.8. These models therefore have a capacity to discriminate on this corpus. Their practical reliability differs, however: GAPL detects 79.9% of AI images but wrongly flags 32.2% of human files, whereas DDA detects 56.6% and wrongly flags 9.8% of human files. Better detection of AI images can thus come at a high cost in false accusations (rows 29, 34, 3, 6).

The other three methods have different limitations. B-Free remains close to the no-discrimination reference in its decisions, with a balanced accuracy of 51.8 without noise, while its AUC reaches only 45.0. PRL achieves a balanced accuracy of 56.9, but its AUC of 49.5 shows no useful overall score ranking between the two origins. Community Forensics achieves a balanced accuracy of 58.9 and an AUC of 66.2, while detecting only 19.5% of AI images. Its low false alert rate of 1.7% therefore coexists with very incomplete detection (rows 29, 34, 3, 6).

With noise, decisions move much closer to the reference of 50: balanced accuracy reaches 56.2 for DDA, 54.5 for SPAI, 52.1 for PRL and 51.3 for GAPL; it falls to 48.6 for Community Forensics and 47.4 for B-Free. The scores retain partial separation for DDA, SPAI and GAPL, with respective AUCs of 62.6, 58.7 and 55.2. In contrast, the AUCs of B-Free, PRL and Community Forensics are 43.1, 44.0 and 35.1: their score ranking is unfavourable to the AI class on these sets. The difficulty therefore does not always arise solely from the decision threshold (rows 29, 34).

These results neither justify treating all six detectors uniformly as equivalent to chance nor establish their general reliability. Some distinguish the two origins under specific conditions and then lose that advantage when the images or their preparation change. Decisions close to chance can also coexist with residual information in

the scores. An isolated verdict, even with a high score, is therefore insufficient to establish an image's origin; the calibration of these scores as probabilities of origin is not evaluated in this study.

Detector robustness and diversity of training data

The results of this study, covering B-Free, SPAI, DDA, PRL, Community Forensics 384 and GAPL, raise a question about the diversity of their training data and, more generally, their ability to recognise an image's origin regardless of the transformations it has undergone. The sometimes considerable changes in detection scores following relatively minor modifications suggest that these models may rely on statistical characteristics that are insufficiently invariant to the conditions in which images are represented.

Preparing reduced-resolution versions of human photographs can increase or decrease their detection score, with a frequently observed tendency towards a decrease. An authentic photograph initially misclassified as AI-generated may thus regain a correct classification after this preparation. The process combines a reduction in resolution, re-encoding and colour profile management; their respective effects are not isolated in this comparison. Adding noise frequently increases scores attributing an artificial origin to human photographs. Its effect on AI images depends on the detector and may instead reduce their detection. These behaviours directly raise questions about the representativeness of the transformations encountered during training and the nature of the characteristics actually learned.

Some methods already incorporate transformations during training: SPAI, for example, uses resizing, Gaussian noise and JPEG compression. It would be relevant to evaluate whether diversifying these transformations and their intensities improves robustness under the conditions studied. Models could be systematically exposed to several versions of the same image, obtained using different resizing operations and interpolation methods, including bicubic or bilinear interpolation, and by adding noise at progressively higher intensities of 1, 2, 5, 10, 20 or 30%, possibly up to 50%, according to a normalised definition of its amplitude. This noise could be introduced independently into the colour channels to produce chromatic perturbations, or applied exclusively to luminance. Changes in brightness, saturation and contrast could also enrich this set. Diversification could extend to RGB and CMYK representations, 8- and 16-bit quantisation depths, and different chroma subsampling schemes: 4:2:0, 4:2:2 and 4:4:4.

In such a protocol, these transformations would be applied symmetrically to human-origin and AI-generated images, consistently preserving their origin labels during training. The objective would be to constrain the models to maintain a consistent classification despite alterations to the signal, reducing their dependence on incidental characteristics associated with resolution, noise, encoding or colour processing.

Such a strategy would not guarantee infallible detection, as some transformations may destroy information needed to identify an image's origin. It would nevertheless provide a way to seek substantially greater robustness than that observed in this study. When a simple technical transformation is enough to radically alter a detector's judgement of an image whose origin remains unchanged, the stability of the detection criterion itself must be questioned.

Corpus and method

B-Free, SPAI, DDA, PRL, Community Forensics 384 and GAPL are evaluated in Sherloq on 666 files. Each detector analysed each file separately, yielding 3,996 usable results. The human photographs are identified as such from their provenance; the AI images were produced with ChatGPT Images 2.5, as used on 3 November 2026.

The corpus comprises 87 original human photographs, their 87 reduced-resolution versions and 87 AI reconstructions produced from these reduced-resolution versions. Each reconstruction reproduces the scene in its reference photograph as faithfully as possible. Another set contains 72 text-generated AI images, prompted to resemble natural photographs of poor quality. All four sets exist without and with noise, giving 333 files per condition: 174 human files and 159 AI files.

The reduction in resolution sets the long edge of the reduced-resolution versions to 1,400 pixels. They were prepared with Photoshop’s Image Processor, whose script uses bicubic interpolation; the files carry an sRGB

profile. This preparation also includes re-encoding. One photograph that was initially smaller was slightly enlarged to reach 1,400 pixels; it remains in the supplied set.

The added noise is Gaussian, with a mean of zero and a standard deviation of 6 levels out of 255, independent for each pixel and each red, green and blue channel. Values are rounded, then clipped between 0 and 255. This operation preserves the dimensions and uses no image generation model. The control illustrated later compares two PNG files prepared in the same way, only one of which receives noise.

Decisions use the recorded native thresholds: a strictly positive logit for B-Free and SPAI; a score of at least 0.5 for DDA and PRL; a score strictly above 0.5 for Community Forensics and GAPL. No threshold was adjusted to the corpus. Scores are shown out of 100; they rank the images according to the model's response.

Reading the indicators

F1 combines AI detection and the proportion of justified alerts. Balanced accuracy gives equal weight to the two classes; with 87 human images and 87 AI images, it equals the proportion of correct decisions. AUC measures the ranking of scores: 50 corresponds to an order with no discrimination, 100 to perfect separation, and a value below 50 to a predominantly reversed order. AP summarises the precision obtained as the threshold varies. IoU, or the Jaccard index, compares the set of flagged images with the set of AI images; it is related to F1 by Jaccard = F1 / (2 - F1), with indices between 0 and 1. For the 87 paired scenes, joint success requires both the human photograph and its AI reconstruction to be correctly classified.

The no-discrimination reference is 50 for both balanced accuracy and AUC, on the scale of 0 to 100 used here. This reference does not apply directly to F1, IoU or AP. For the raw proportion of correct decisions, a random decision giving both labels an equal chance achieves 50% on average; an always "human" decision achieves 52.3% on the 333 files in each condition, which comprise 174 human files and 159 AI files.

The "All categories combined" rows bring together the four sets in each condition. Each file counts once, including both human versions. Overall indicators are recalculated from the pooled decisions and scores.

In the table, noise applies to all the sets named in the row. Percentages carry the % sign; scores and indices are out of 100. The proportion of justified alerts concerns the flagged images; correct decisions concern all the files in question. n/a means that no image is flagged. The number of changed verdicts appears only on the side with noise.

L.	Images concerned	Without noise						With noise					
		B-Free	SPAI	DDA	PRL	Community Forensics 384	GAPL	B-Free	SPAI	DDA	PRL	Community Forensics 384	GAPL
	Among AI-generated images, which are detected?												
01	AI reconstruction	6.9%	70.1%	63.2%	42.5%	24.1%	96.6%	3.4%	85.1%	34.5%	25.3%	3.4%	49.4%
02	AI prompt	11.1%	52.8%	48.6%	6.9%	13.9%	59.7%	8.3%	54.2%	38.9%	5.6%	0.0%	48.6%
03	**All categories combined**	**8.8%**	**62.3%**	**56.6%**	**26.4%**	**19.5%**	**79.9%**	**5.7%**	**71.1%**	**36.5%**	**16.4%**	**1.9%**	**49.1%**
	Among human photographs, which are wrongly flagged?												
04	Original photo	8.0%	33.3%	12.6%	8.0%	3.4%	52.9%	16.1%	63.2%	40.2%	8.0%	9.2%	70.1%
05	Reduced version	2.3%	26.4%	6.9%	17.2%	0.0%	11.5%	5.7%	60.9%	8.0%	16.1%	0.0%	23.0%
06	**All categories combined**	**5.2%**	**29.9%**	**9.8%**	**12.6%**	**1.7%**	**32.2%**	**10.9%**	**62.1%**	**24.1%**	**12.1%**	**4.6%**	**46.6%**
	Median score (out of 100)												
07	Original photo	15.6	0.1	17.5	21.5	0.1	58.5	20.7	94.9	36.7	21.6	1.3	81.8
08	Reduced version	2.3	0.0	7.2	26.0	0.1	0.8	3.4	81.6	8.0	24.6	0.1	8.4
09	AI reconstruction	2.7	98.4	86.2	43.2	2.6	100.0	4.6	100.0	29.1	33.2	0.1	49.9
10	AI prompt	9.4	60.0	45.5	15.8	0.1	81.7	11.5	58.8	41.6	15.7	0.0	48.2
11	**All categories combined**	**6.4**	**14.1**	**23.7**	**23.5**	**0.3**	**67.7**	**9.2**	**94.3**	**25.4**	**22.0**	**0.1**	**44.8**
	Among alerts, which are justified?												
12	Original photo and AI reconstruction	46.2%	67.8%	83.3%	84.1%	87.5%	64.6%	17.6%	57.4%	46.2%	75.9%	27.3%	41.3%
13	Reduced version and AI reconstruction	75.0%	72.6%	90.2%	71.2%	100.0%	89.4%	37.5%	58.3%	81.1%	61.1%	100.0%	68.3%
14	Original photo and AI prompt	53.3%	56.7%	76.1%	41.7%	76.9%	48.3%	30.0%	41.5%	44.4%	36.4%	0.0%	36.5%
15	Reduced version and AI prompt	80.0%	62.3%	85.4%	25.0%	100.0%	81.1%	54.5%	42.4%	80.0%	22.2%	n/a	63.6%
16	**All categories combined**	**60.9%**	**65.6%**	**84.1%**	**65.6%**	**91.2%**	**69.4%**	**32.1%**	**51.1%**	**58.0%**	**55.3%**	**27.3%**	**49.1%**
	Across human and AI images, which decisions are correct?												
17	Original photo and AI reconstruction	49.4%	68.4%	75.3%	67.2%	60.3%	71.8%	43.7%	60.9%	47.1%	58.6%	47.1%	39.7%
18	Reduced version and AI reconstruction	52.3%	71.8%	78.2%	62.6%	62.1%	92.5%	48.9%	62.1%	63.2%	54.6%	51.7%	63.2%
19	Original photo and AI prompt	55.3%	60.4%	69.8%	53.5%	59.1%	52.8%	49.7%	44.7%	50.3%	52.8%	49.7%	38.4%
20	Reduced version and AI prompt	58.5%	64.2%	73.0%	48.4%	61.0%	75.5%	55.3%	45.9%	67.9%	48.4%	54.7%	64.2%
21	**All categories combined**	**53.8%**	**66.4%**	**74.2%**	**58.3%**	**60.7%**	**73.6%**	**49.2%**	**53.8%**	**57.1%**	**53.8%**	**50.8%**	**51.4%**
	AI-class F1 (out of 100)												
22	Original photo and AI reconstruction	**12.0**	**68.9**	**71.9**	**56.5**	**37.8**	**77.4**	**5.8**	**68.5**	**39.5**	**37.9**	**6.1**	**45.0**
23	Reduced version and AI reconstruction	**12.6**	**71.3**	**74.3**	**53.2**	**38.9**	**92.8**	**6.3**	**69.2**	**48.4**	**35.8**	**6.7**	**57.3**
24	Original photo and AI prompt	**18.4**	**54.7**	**59.3**	**11.9**	**23.5**	**53.4**	**13.0**	**47.0**	**41.5**	**9.6**	**0.0**	**41.7**
25	Reduced version and AI prompt	**19.5**	**57.1**	**61.9**	**10.9**	**24.4**	**68.8**	**14.5**	**47.6**	**52.3**	**8.9**	**0.0**	**55.1**
26	**All categories combined**	**15.4**	**63.9**	**67.7**	**37.7**	**32.1**	**74.3**	**9.6**	**59.5**	**44.8**	**25.2**	**3.5**	**49.1**
	Balanced accuracy (out of 100)												
27	Original photo and AI prompt	**51.5**	**59.7**	**68.0**	**49.4**	**55.2**	**53.4**	**46.1**	**45.5**	**49.3**	**48.8**	**45.4**	**39.2**
28	Reduced version and AI prompt	**54.4**	**63.2**	**70.9**	**44.9**	**56.9**	**74.1**	**51.3**	**46.6**	**65.4**	**44.7**	**50.0**	**62.8**
29	**All categories combined**	**51.8**	**66.2**	**73.4**	**56.9**	**58.9**	**73.8**	**47.4**	**54.5**	**56.2**	**52.1**	**48.6**	**51.3**
	AUC: score ranking (out of 100)												
30	Original photo and AI reconstruction	**25.9**	**74.8**	**85.5**	**77.6**	**79.2**	**89.7**	**22.4**	**68.8**	**45.0**	**66.5**	**31.0**	**41.3**
31	Reduced version and AI reconstruction	**53.4**	**78.2**	**92.9**	**69.1**	**77.9**	**98.3**	**52.3**	**74.0**	**77.2**	**59.5**	**55.6**	**71.4**
32	Original photo and AI prompt	**38.9**	**65.3**	**65.7**	**23.3**	**51.5**	**60.2**	**36.4**	**40.5**	**48.7**	**22.5**	**15.9**	**36.6**
33	Reduced version and AI prompt	**64.0**	**71.3**	**79.0**	**18.1**	**51.2**	**84.8**	**63.8**	**46.4**	**80.3**	**19.8**	**34.5**	**71.1**
34	**All categories combined**	**45.0**	**72.8**	**81.6**	**49.5**	**66.2**	**84.3**	**43.1**	**58.7**	**62.6**	**44.0**	**35.1**	**55.2**
	AP: average precision (out of 100)												
35	Original photo and AI reconstruction	**39.2**	**77.4**	**87.7**	**78.6**	**79.2**	**90.1**	**36.9**	**69.9**	**52.0**	**68.8**	**39.0**	**49.0**
36	Reduced version and AI reconstruction	**55.0**	**75.8**	**92.7**	**71.5**	**79.6**	**98.5**	**52.3**	**74.1**	**75.4**	**62.7**	**55.9**	**73.9**
37	Original photo and AI prompt	**42.3**	**56.0**	**72.3**	**34.7**	**58.7**	**61.7**	**38.4**	**37.6**	**47.4**	**34.0**	**29.8**	**38.5**
38	Reduced version and AI prompt	**60.6**	**55.8**	**79.7**	**31.4**	**59.1**	**84.7**	**57.8**	**41.1**	**74.5**	**32.7**	**36.7**	**66.6**
39	**All categories combined**	**46.7**	**68.1**	**83.8**	**56.7**	**70.4**	**85.1**	**43.5**	**57.8**	**59.2**	**50.6**	**38.4**	**53.3**
	AI-class IoU / Jaccard (out of 100)												
40	Original photo and AI reconstruction	**6.4**	**52.6**	**56.1**	**39.4**	**23.3**	**63.2**	**3.0**	**52.1**	**24.6**	**23.4**	**3.2**	**29.1**
41	Reduced version and AI reconstruction	**6.7**	**55.5**	**59.1**	**36.3**	**24.1**	**86.6**	**3.3**	**52.9**	**31.9**	**21.8**	**3.4**	**40.2**
42	Original photo and AI prompt	**10.1**	**37.6**	**42.2**	**6.3**	**13.3**	**36.4**	**7.0**	**30.7**	**26.2**	**5.1**	**0.0**	**26.3**
43	Reduced version and AI prompt	**10.8**	**40.0**	**44.9**	**5.7**	**13.9**	**52.4**	**7.8**	**31.2**	**35.4**	**4.7**	**0.0**	**38.0**
44	**All categories combined**	**8.3**	**46.9**	**51.1**	**23.2**	**19.1**	**59.1**	**5.1**	**42.3**	**28.9**	**14.4**	**1.8**	**32.5**
	Scenes with both decisions correct (%)												
45	Original photo and AI reconstruction	3.4%	44.8%	55.2%	37.9%	21.8%	44.8%	1.1%	26.4%	18.4%	20.7%	1.1%	13.8%
46	Reduced version and AI reconstruction	5.7%	44.8%	57.5%	32.2%	24.1%	85.1%	1.1%	25.3%	27.6%	18.4%	3.4%	31.0%
	AI score higher than human score (%)												
47	Original photo and AI reconstruction	28.7%	78.2%	85.1%	80.5%	87.4%	90.8%	16.1%	70.1%	37.9%	74.7%	23.0%	37.9%
48	Reduced version and AI reconstruction	55.2%	93.1%	95.4%	67.8%	89.7%	100.0%	54.0%	90.8%	81.6%	56.3%	64.4%	77.0%
	Both images classified as AI (scenes)												
49	Reduced version and AI reconstruction	1	22	5	9	0	10	2	52	6	6	0	16
	Both images classified as non-AI (scenes)												
50	Reduced version and AI reconstruction	80	25	31	44	66	3	81	12	56	57	84	40
	Human photo wrongly flagged and AI image missed (scenes)												
51	Reduced version and AI reconstruction	1	1	1	6	0	0	3	1	1	8	0	4
	Decisions changed by noise (images)												
52	Original photo	-	-	-	-	-	-	7	28	28	0	7	33
53	Reduced version	-	-	-	-	-	-	3	30	5	7	0	20
54	AI reconstruction	-	-	-	-	-	-	3	15	29	19	18	41
55	AI prompt	-	-	-	-	-	-	8	9	23	1	10	24
56	**All categories combined**	**-**	**-**	**-**	**-**	**-**	**-**	**21**	**82**	**85**	**27**	**35**	**118**
	Reconstructions missed by both the model and GAPL (images)												
57	AI reconstruction	3	1	1	3	2	-	41	4	33	33	41	-
	Detected here and missed by GAPL (images)												
58	AI reconstruction	0	2	2	0	1	-	3	40	11	11	3	-
	Detected by GAPL and missed here (images)												
59	AI reconstruction	78	25	31	47	64	-	43	9	24	32	43	-

Results without noise by image type

Across all categories, DDA achieves 74.2% correct decisions without noise, ahead of GAPL at 73.6%. GAPL detects more AI images, 79.9% versus 56.6%, but wrongly flags 32.2% of human files versus 9.8% for DDA. Its overall F1 remains higher, at 74.3 versus 67.7. After noise is added, SPAI achieves the best overall F1, at 59.5; DDA retains the highest proportion of correct decisions, at 57.1%, versus 53.8% for SPAI. The ranking therefore depends on the weight given to AI detection and wrongly flagged human images (rows 21, 3, 6, 26).

In the set of 87 reduced-resolution human photographs and 87 AI reconstructions without noise, B-Free correctly recognises 85 human photographs and detects 6 AI images. This gives 91 correct decisions among the 174 images examined, or 52.3%. On the same images after noise is added, it correctly recognises 85 in total, or 48.9%. (rows 18).

GAPL detects 96.6% of the AI reconstructions, ahead of SPAI at 70.1% and DDA at 63.2%. PRL detects 42.5%, Community Forensics 24.1% and B-Free 6.9%. The best detection result is therefore almost exhaustive, while the weakest misses 81 reconstructions out of 87. Combined with the reduced-resolution human photographs, GAPL also achieves the best F1, at 92.8, and 92.5% correct decisions. These results are brought together in rows 1, 23 and 18.

This hierarchy deteriorates on text-generated AI images. GAPL falls to 59.7% of AI images detected, SPAI to 52.8% and DDA to 48.6%. PRL's decline is particularly sharp: from 42.5% on the reconstructions to 6.9% on text-generated AI images. Community Forensics falls to 13.9%. B-Free improves slightly, to 11.1%, but still misses 64 images out of 72. Five methods out of six therefore detect this second set less well, with losses ranging from 10.2 to 36.8 points (rows 1-2). The type of generated image markedly changes performance, even with the same generation source.

False alerts change the practical ranking. On the reduced-resolution versions, DDA wrongly flags 6.9% of human photographs, GAPL 11.5% and SPAI 26.4%. SPAI detects more reconstructions than DDA, but wrongly flags almost four times as many human photographs. DDA then achieves an F1 of 74.3 versus 71.3 for SPAI, and 78.2% correct decisions versus 71.8%. Community Forensics produces no false alerts on the reduced-resolution versions; its 100% justified alerts correspond to only 21 reconstructions detected out of 87, or 10 text-generated AI images out of 72. The reliability of its alerts coexists here with low coverage of AI images (rows 5, 13, 15).

The text-generated AI image set is particularly unfavourable to PRL. When this set is combined with the reduced-resolution photographs, PRL wrongly flags 15 human photographs and detects only 5 AI images: one alert in four is justified, and its balanced accuracy is 44.9. Its AUC of 18.1 shows that the score itself ranks these two sets very poorly. The problem therefore goes beyond the choice of threshold alone (rows 15, 28, 33).

Comparison of original and reduced-resolution photographs

GAPL stands out in the comparison between originals and reduced-resolution versions. Without noise, its false alerts fall from 52.9% to 11.5%, a decrease of 41.4 points. The median of its human-image scores falls from 58.5 to 0.8 out of 100. The AI reconstructions remain exactly the same in this comparison: replacing the human photographs is what raises correct decisions from 71.8% to 92.5%. With text-generated AI images, the same replacement raises this rate from 52.8% to 75.5% (rows 4-5, 7-8, 17-20).

GAPL thus performs very well on reconstructions compared with reduced-resolution photographs, while wrongly flagging many originals. Its sensitivity to file preparation directly limits its usefulness for examining photographs of varied formats and sizes. DDA shows a more moderate effect: false alerts decrease from 12.6% to 6.9%. B-Free falls from 8.0% to 2.3%, SPAI from 33.3% to 26.4% and Community Forensics from 3.4% to 0%. PRL moves in the opposite direction, from 8.0% to 17.2% (rows 4-5). The reduction in resolution therefore does not bring a benefit shared by all six methods.

With noise, this gap between formats becomes even larger for DDA: it wrongly flags 40.2% of originals, versus 8.0% of reduced-resolution versions. For GAPL, the rates reach 70.1% and 23.0% respectively. SPAI still wrongly flags many photographs in both cases, at 63.2% and 60.9%. Both forms of the human set are needed to characterise these behaviours. The Photoshop processing changes resolution, encoding and profile together; the observed differences concern this complete preparation.

Effect of noise on human photographs

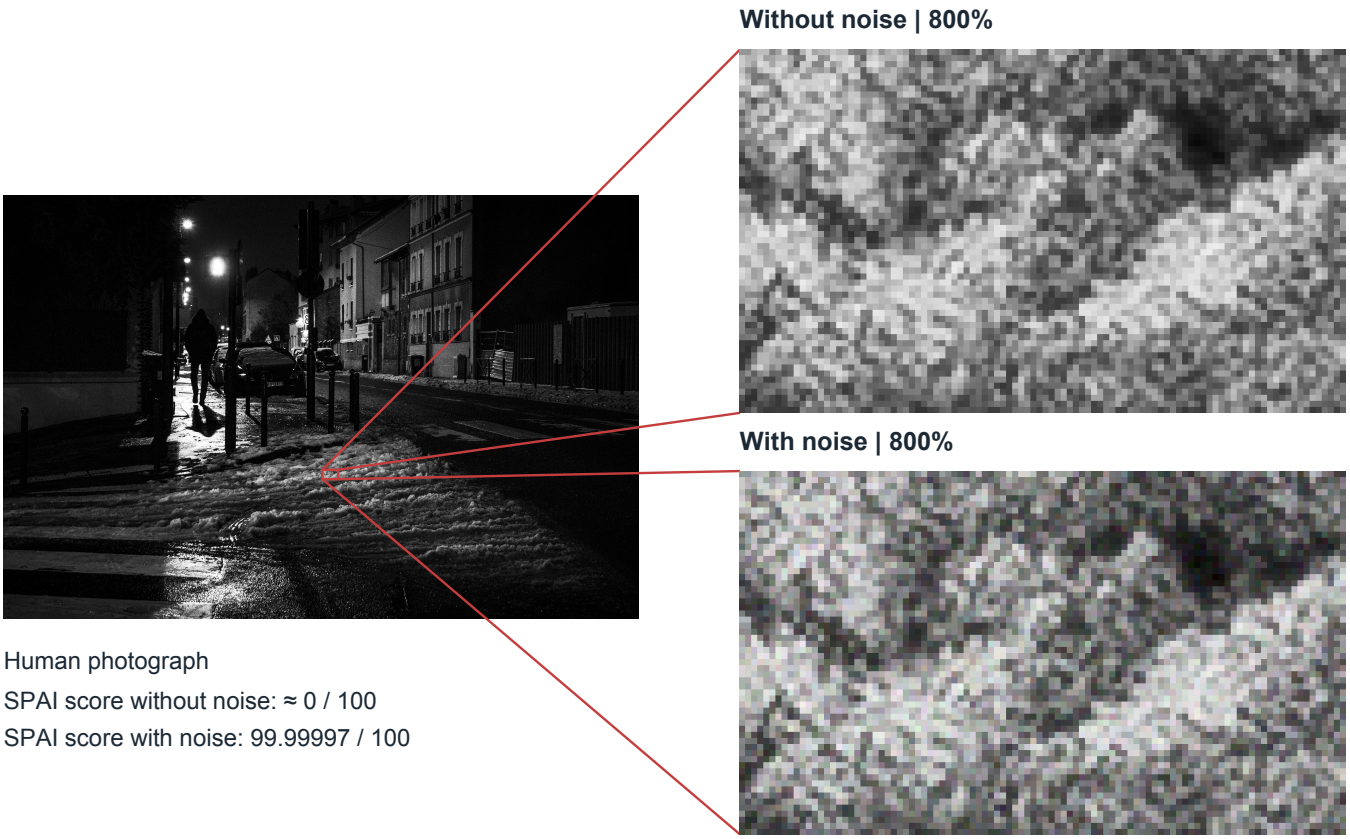
On the originals, SPAI rises from 33.3% to 63.2% false alerts: 55 human photographs out of 87 are declared AI. DDA rises from 12.6% to 40.2%, or 11 then 35 wrongly flagged photographs. Its number of errors increases by a factor of 3.2. GAPL, already struggling at 52.9%, rises to 70.1%. B-Free rises from 8.0% to 16.1% and Community Forensics from 3.4% to 9.2%. PRL retains exactly the same decisions on the 87 originals (rows 4, 52).

On the reduced-resolution versions, SPAI's response remains very strong: from 26.4% to 60.9% false alerts. Thirty photographs previously recognised correctly are now wrongly flagged; none moves in the opposite direction. GAPL doubles its rate, from 11.5% to 23.0%. DDA rises only from 6.9% to 8.0%, while Community Forensics remains at zero. PRL falls slightly, from 17.2% to 16.1%, with seven changed verdicts that largely offset one another (rows 5, 53). DDA therefore withstands noise much better on the reduced-resolution versions than on the originals.

SPAI's scores shift massively: the median for the human originals rises from 0.1 to 94.9 out of 100, and that for the reduced-resolution photographs from 0.0 to 81.6. These values are median scores; the proportions of wrongly flagged photographs are 63.2% and 60.9% respectively. For GAPL, the reduced-resolution versions retain a median of 8.4 after noise is added, while producing 23.0% false alerts. A low median can therefore still leave a sizeable group of images above the threshold (rows 7-8, 4-5).

The snowy photograph below shows SPAI switching its verdict in a control that preserves dimensions, decoding and PNG encoding. The file without noise receives exactly the same score as the original photograph, about 0.0000000006 out of 100. After Gaussian noise with a standard deviation of 6 is added, the score reaches 99.99997 out of 100 and the verdict becomes AI. The change is produced by numerically adding noise to the human photograph.

SPAI's oversensitivity to noise



Original photograph and the same snow region at 800%, without noise then with noise. Scores concern the entire 4,034 × 2,689-pixel images. In this control, the score without noise is identical to that of the original file; the second file receives only the numerical addition of noise after the same PNG preparation.

Effect of noise on AI-generated images

Noise causes GAPL to lose 41 reconstruction detections, with no gains: from 84 images detected to 43, or from 96.6% to 49.4%. DDA falls from 63.2% to 34.5% and PRL from 42.5% to 25.3%. Community Forensics loses 18 of its 21 detections, ending at 3.4%, the same level as B-Free. SPAI moves in the opposite direction: from 70.1% to 85.1% (rows 1, 54).

This rise in SPAI is accompanied, however, by the wrongly flagged human photographs described above. When reduced-resolution photographs and reconstructions are pooled, the proportion of justified alerts falls from 72.6% to 58.3%, and correct decisions from 71.8% to 62.1%. SPAI responds more in favour of the AI class for both origins. Its detection gain is insufficient to offset the new false alerts. Its F1 changes little, from 71.3 to 69.2, while its correct decisions lose 9.8 points (rows 13, 18, 23).

On text-generated AI images, Community Forensics no longer detects any of the 72 images after noise is added. B-Free and PRL detect only 6 and 4. DDA drops from 35 to 28 detections, GAPL from 43 to 35; SPAI remains almost stable, from 38 to 39. Underlying this overall stability for SPAI are nine verdict changes: four detections lost and five gained. GAPL loses sixteen and gains eight, for a net loss of eight detections (rows 2, 55).

The overall result still depends on the human set. With the reduced-resolution versions and noisy text-generated AI images, DDA retains 67.9% correct decisions, a balanced accuracy of 65.4 and 80.0% justified alerts. With the noisy originals, the same 28 detected AI images are accompanied by 35 wrongly flagged human photographs: the proportion of justified alerts falls to 44.4% and correct decisions to 50.3%. For GAPL, accuracy on noisy originals and text-generated AI images falls to 38.4%. Behaviour on human photographs can therefore account for most of the final failure (rows 20, 28, 15, 14, 19).

Score ranking and class separation

Without noise, DDA separates the reduced-resolution versions of human photographs from AI reconstructions very well: its AUC reaches 92.9, whereas the native threshold detects only 63.2% of the AI reconstructions. GAPL achieves an AUC of 98.3 and a detection rate of 96.6%. Community Forensics achieves an AUC of 77.9 while detecting only 24.1% of the AI images. DDA and Community Forensics scores therefore contain a separation that their decisions at the native threshold exploit only partly (rows 31, 1).

After noise is added, the ranking itself can collapse. With the original human photographs and AI reconstructions, DDA's AUC falls from 85.5 to 45.0, Community Forensics' from 79.2 to 31.0, and GAPL's from 89.7 to 41.3. The AI reconstructions then frequently receive lower scores than the human photographs. A single shift in the threshold cannot correct this inverted order. With the reduced-resolution versions, DDA and GAPL retain more useful separation, at 77.2 and 71.4 (rows 30-31).

For text-generated images with added noise, DDA achieves an AUC of 80.3 with the reduced-resolution versions, compared with 48.7 with the originals. SPAI falls from 71.3 to 46.4 with the reduced-resolution versions; Community Forensics falls from 51.2 to 34.5. PRL already performs very poorly in terms of score orientation before noise is added, and its AUC remains close to 20 afterwards (rows 32-33). AP confirms several of these losses: with reduced-resolution versions of human photographs and AI reconstructions, GAPL falls from 98.5 to 73.9, DDA from 92.7 to 75.4, and Community Forensics from 79.6 to 55.9 (rows 36).

Paired comparisons and shared errors

The matches between human photographs and AI reconstructions allow a stronger requirement: correctly recognising both images of the same scene. Without noise, GAPL achieves this for 85.1% of pairs consisting of a reduced-resolution version and an AI reconstruction, DDA for 57.5%, SPAI for 44.8%, PRL for 32.2%, Community Forensics for 24.1%, and B-Free for 5.7%. With the original human photographs, GAPL falls to 44.8%, while DDA reaches 55.2%. GAPL's high detection rate for AI reconstructions therefore still leaves many scenes whose human photograph is also flagged as AI (rows 45, 46).

After noise is added, GAPL gets both decisions right for only 31.0% of pairs using reduced-resolution versions and 13.8% of those using originals. DDA falls to 27.6% and 18.4%, respectively. For SPAI, both images are classified as AI in 52 out of 87 pairs consisting of a reduced-resolution version and an AI reconstruction, compared with 22 before noise is added. Community Forensics, by contrast, classifies both images as non-AI in 84 out of 87 pairs. The failures therefore divide into assigning the AI label to both origins and removing it from both (rows 45, 46, 49-50).

The score order helps explain some of these errors. Without noise, GAPL gives the AI reconstruction a higher score than the reduced-resolution version of the corresponding human photograph in all 87 scenes. DDA does so in 95.4% of scenes, SPAI in 93.1%, and Community Forensics in 89.7%. After noise is added, SPAI retains this order in 90.8% of pairs, but correctly classifies both images in only 25.3% of cases. The relative ranking can remain good while both scores cross the AI threshold together (rows 48, 46).

The methods do not always miss the same images. Of the 44 AI reconstructions with added noise that GAPL misses, SPAI detects 40 and misses 4. Conversely, GAPL detects 9 AI reconstructions that SPAI misses. DDA and PRL each detect 11 of the AI reconstructions missed by GAPL, compared with 3 each for B-Free and Community Forensics. These complementarities can be measured on the files themselves; they coexist with very different false-alert rates (rows 57-59, 4-5).

Uncertainty and scope of the study

The differences described concern all files in the corpus. For changes in the decision on the same image, the six planned comparisons for each of the six detectors are also examined using an exact McNemar test, with Holm correction across the 36 tests. The rise in SPAI's false alerts on human photographs, the rise for DDA on the originals, and the losses in AI reconstruction detection for DDA, PRL, Community Forensics and GAPL remain significant at the 5% level. GAPL's 17.2-point rise in false alerts on the originals does not reach this level after correction: $p = 0.311$, above the chosen threshold of 0.05. The 95% intervals for the changes, calculated from 5,000 resamplings of the families, and the test results are supplied with the numerical data.

Comparisons of balanced accuracy and AUC against the reference value of 50 are descriptive. The McNemar tests and the intervals for the changes cited above assess changes in decisions; they are not tests of the departure of performance from chance.

Limitations and representativeness of the evaluation corpus

The corpus used in this study has several representativeness biases. The human photographs come from a personal collection assembled mainly between 2018 and 2019, almost exclusively with a DSC SONY RX10. They mainly reflect French environments and a Western photographic perspective, with a predominance of urban landscapes, structures, textures and natural features. People and animals are relatively underrepresented, while many geographical and cultural environments are absent. This homogeneity in equipment, time period and subject matter limits the generalisation of the results.

The selected subjects also largely belong to visual categories that are probably well represented in the training data of image generators. Rarer or less well-documented subjects could lead to more anatomical, structural or contextual inconsistencies during generation. However, no direct relationship can be established between the frequency of these perceptual anomalies and the presence of statistical signatures actually used by the detectors. It therefore remains impossible to determine whether diversifying the corpus to include subjects that are more difficult to generate would have made automated detection easier.

The original human photographs also frequently have characteristics far removed from idealised artificial representations: blur, noise, low light, dull colours, unusual framing and unconventional compositions. Reproducing them with artificial intelligence may have weakened some aesthetic cues usually associated with synthetic images, potentially making discrimination more demanding.

The wording of the generation instructions is another source of uncertainty. By explicitly requesting images that are difficult to distinguish from human photographs, the protocol could steer the generator towards representations that weaken some visual cues usually associated with artificial outputs. It is even conceivable that, during its own training, the generative model learnt some characteristics that distinguish synthetic images from human photographs, and that this statistical knowledge indirectly influences its outputs when a realistic photographic appearance is requested. Such a hypothesis would nevertheless require these characteristics to be represented in the model and to be modulated through the generation instructions. No documentary or experimental evidence available in this study can confirm this or measure its possible contribution to the results. This possibility must therefore be regarded as a methodological uncertainty rather than an established mechanism.

The almost exclusive use of the same camera could introduce recurring technical signatures into the human photographs, potentially influencing classification independently of their origin. The AI images come from a single generation source; some prompts produced multiple images. Variants derived from the same source are related, and the 666 files do not correspond to 666 independent scenes. AI reconstructions and text-generated images also differ in their content and instructions. These factors, whose effects may conflict, require the interpretation of the observed performance to be limited to the particular conditions of the corpus, protocol and implementation evaluated in Sherloq.

The reduction in resolution combines a change in pixel dimensions, re-encoding and profile handling. The copies with added noise are PNG files, while the supplied human photographs are JPEG, MPO or TIFF files. Their preparation therefore also involves format conversion; the control using the photograph of a snowy scene retains a common PNG encoding and isolates the addition of noise in that case. Removing the slightly enlarged photograph preserves the decrease in GAPL's false alerts between the originals and reduced-resolution versions. The individual results, error counts and numerical comparisons remain available in the attached data files.

The role of artificial intelligence

The corpus's 87 AI reconstructions and 72 text-generated images were produced with ChatGPT Images 2.5. The reference human photographs and their reduced-resolution versions or versions with added noise were prepared without a generative model; the reduction in resolution and addition of noise use conventional digital processing. The six detectors evaluated are themselves artificial intelligence models.

An AI assistant helped translate the protocol into code, execute it, organise the results, and calculate metrics and statistical comparisons. It contributed to numerical checks, the construction of the central table and the formatting of the document. Its assistance also covered consulting scientific publications, proposing analyses and interpretations, and drafting and rephrasing the different sections. The reported values come from programmed calculations applied to the detectors' recorded results.

Human work covered assembling the corpus, selecting relevant questions and comparisons, critically interpreting the results and examining the limitations. It particularly concerned the origin and representativeness of the source images, the conditions of their production and the epistemic quality of the material studied. The AI's proposals were selected, corrected or discarded on the basis of this analysis; the interpretations retained and the scientific direction reflect human choices.

Method references

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